



Sunil

Roll No.: 230101099

B. Tech. - Computer Science and Engineering

Indian Institute of Technology, Guwahati

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## EDUCATION

| Degree/Certificate | Institute/Board                          | CGPA/Percentage | Year         |
|--------------------|------------------------------------------|-----------------|--------------|
| B.Tech.            | Indian Institute of Technology, Guwahati | 9.15 (Current)  | 2023-Present |
| Senior Secondary   | JNV Alwar ( CBSE )                       | 96.8%           | 2022         |
| Secondary          | JNV Alwar ( CBSE )                       | 96.4%           | 2020         |

## PROJECTS

- **MIPS Processor** April 2025  
*Course Project of Hardware Lab under Dr. Chandan Karfa* Source-Code
  - Designed a **Verilog-based** single-cycle MIPS CPU integrating a 32-bit ALU, register file, instruction and data memory, program counter, and centralized control logic for precise instruction execution and flow control.
  - Implemented **arithmetic** (ADD, SUB, SLT), **logic** (AND, OR), **memory** (LW, SW), and **control flow** (BEQ, BNE, J) instructions with structured opcode decoding and a fully functional, **modular datapath** architecture.
  - Validated design correctness via **hex-based testbenches** and  $\$display()$  tracing in ModelSim, ensuring accurate instruction sequencing, memory updates, branch resolution, and control signal behavior.
- **CompanyGrow - Smart Workforce Development Platform** May 2025 - June 2025  
*Coding Club, IIT Guwahati* Source-Code | Live Demo
  - Developed a scalable **full-stack platform** to drive employee growth through structured training programs, intelligent skill-based project allocation, personalized learning paths, and performance-based rewards and recognition.
  - Designed a course catalog with skill filters and enrollment features using **React(JavaScript)** and implemented secure, **JWT-based authentication** and a smart skill-matching engine to optimize project assignments. .
  - Integrated **Stripe APIs** for real-time badge-based rewards, utilised **Cloudinary** for media and **Chart.js** for analytics and exported detailed progress reports (pdf) using **pdf-lib & jsPDF**, supporting seamless reporting.
- **Inventory Optimization using Advanced SQL Analytics** June 2025  
*Consulting & Analytics Club, IIT Guwahati* Source-Code
  - Designed and implemented a scalable **SQL analytics framework** for Urban Retail Co., enabling data-driven resolution of inventory inefficiencies across **5,000+ SKUs** and multiple retail regions.
  - Built dynamic dashboards and crafted complex queries to monitor stock health, optimize reorder points, track inventory turnover, uncover seasonal demand trends, and evaluate promotion effectiveness.
  - Utilized advanced SQL techniques—**window functions**, **CTEs**, **CASE logic**, **aggregations**, and **schema normalization** to develop high-performance, maintainable queries and deliver concise, executive-ready **KPI dashboards**.
- **Zcoder – Interactive Code Practice with Live Collaboration** May 2025 - June 2025  
*Coding Club , IIT Guwahati* Source-Code
  - Developed a **dynamic full-stack platform** for personalized coding practice featuring real-time feedback, intelligent progress tracking, and peer-to-peer collaboration to cultivate a community-driven learning ecosystem.
  - Empowered users with advanced features such as problem bookmarking, **topic tagging for structured learning**, and detailed user profiles showcasing **progress analytics**, activity history, and performance insights.
  - Integrated a real-time **code editor** and **collaborative rooms** using **WebSockets**, enabling seamless pair programming.

## ACHIEVEMENTS

- **Reliance Foundation UG scholarship**, Awarded for academic excellence and leadership potential. 2023-2027
- **Codeforces**, Maximum Rating - **1466** | Specialist | Handle - **typocode**
  - Secured **Global Rank 631** in Codeforces Round 1027 (Div. 3) among 38k+ participants.
- **Graph Camp by Manas Sir**, AlgoUniversity - learnt **17+** advanced problem-solving techniques. April 2025
- **JEE Advanced** ,Secured **AIR 1369** among 0.25 million candidates appearing for the test 2023
- **JEE Mains** ,Secured **AIR 2848** among 1.2 million candidates appearing for the test 2023

## TECHNICAL SKILLS

- **Programming:** C/C++, Python\*
- **Web Development:** HTML, CSS, React, Nodejs\*
- **Frameworks:** NumPy, Pandas\*
- **Miscellaneous:** MySQL, Windows, Linux, Git, Verilog

\* Elementary proficiency

## KEY COURSES TAKEN

- **Computer Science:** Algorithms and Data Structures\*, Digital Design, Design and Analysis of Algorithms\*, Computer Architecture and Organization\*, Database Management Systems\*
- **Mathematics:** Probability Theory and Random Processes, Elementary Number Theory and Algebra, Discrete Mathematics, Linear Algebra \*AA grade

## EXTRACURRICULAR ACTIVITIES

- **PR & Marketing Executive**, Red Ribbon Club, IIT Guwahati April 2024 - Present
  - Spearheaded promotion and outreach for **Cyclothon** and **Blood Donation Camp** during annual welfare fest **Prayatna**, that attracted over **300+** participants for Cyclothon and helped collect **150+ units** of Blood.
- **Silver Medalist**, Volleyball - Spardha , Inter-Hostel Sports Competition, IIT Guwahati Feb 2024